

Spatiotemporal-Memory-Guided Machine Perception for Augmented Reality

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ABSTRACT

Augmented Reality (AR) devices are equipped with advanced sensors such as panoramic cameras. They can see or hear what humans cannot. The superior sensing capabilities - together with machine perception models - enable promising AR-based human assistants in daily tasks. Human task performance progressively enhances with accumulated memory, such as with observed objects' location in working memory, and training in long-term memory. We transform this cognitive nature into real-time machine perception models tailored for human-assisting AR applications. Instead of running machine perception models on each consecutive frame, we leverage the egocentric AR cameras to reconstruct a "4D memory map", which mimics the human spatiotemporal memory.

Index Terms: Perception and cognition; Modeling and Simulation

1 INTRODUCTION

Unlike Virtual Reality (VR), Augmented Reality (AR) requires not only high-quality imagery, but real-time environment perception, such as surface tracking and object tracking. Deep learning is a broadly enabling technology for those machine perception computations. However, deep learning models commonly demand high computational resources, limiting their real-time applicability on low-power edge devices. Recent advances in visual perception system is further driven by deeper and larger neural networks for optimal accuracy, further compromising practical usage. For example, the popular AR device, Microsoft Hololens 2, is equipped only with a mobile platform processor. Consequently, real-time machine perception may fail due to the high latency from the demanding neural network inference [2]. One promising solution is to leverage cloud-based frameworks by performing the demanding computation on the server and streaming the results to the edge AR devices [4]. However, the network transmission bandwidth may inevitably harm spatial accuracy or temporal latency [5]. The bottom left of Fig. 1 shows an example of real-time perception. In such settings, the cloud may perform computation on a past frame, and mistakenly return an outdated object location to the AR device. A similar challenge has been observed for autonomous vehicle perception [3].

To develop AR-tailored, scalable, and real-time machine perception while preserving high accuracy and speed, we introduce and integrate a spatial memory module into the machine learning framework. Our method is inspired by the interplay between two forms of human declarative memories - recognition and location [1]. We propose SpatioTEMporal-memory-guided machine perception for AR (STEMAR). Instead of limiting the computation independently to users' current field of view (Fov), STEMAR memorizes the dynamic environments (The right of Fig. 1) to create a spatiotemporal 4D knowledge of the physical environment, and in turn, exploits the memory to avoid the repeated perception of unchanged envi-

ronments for perception acceleration. In addition, STEMAR works in hypothesis testing to verify the objects in small local patches instead of detecting objects from the whole frame, resulting in higher detection accuracy.

With the memorized 4D knowledge, STEMAR is capable of linking objects after a long time span and identifying changes in both appearances and locations of objects, which cannot be achieved in computer vision (CV) algorithms (detection, tracking, person re-identification). Our quantitative measurement and qualitative visualization demonstrate STEMAR's major advantages, including, significantly enhanced AR on-device object tracking speed (by 73%), accuracy (by 34.6%), and robustness (by 23%), as well as extended spatial (indicating task objects beyond current FoV) and temporal (identifying objects' location changes from the past) span for practical user-assisting machine vision models in AR.

2 STEMAR - THE PROPOSED FRAMEWORK

2.1 AR Hardware

HoloLens2 is equipped with high quality sensors. Beyond an RGB camera, also known as PV camera (1280 × 720, 15/30fps), and microphone, HoloLens2 has a research mode providing access to extra sensors. The sensors that can be unlocked by research mode include 4 grayscale cameras used by the system for head tracking and map building, the infrared camera which can capture the depth and active brightness, an inertial measurement unit (IMU) which contains an accelerometer, gyro, and magnetometer. HoloLens2 can also provide eye tracking and hand tracking results.

2.2 Method

The general framework built on Hololens2 includes three major parts: perception initialization, transformation, memorization and re-identification. A general flowchart can be found in Fig. 1. Hololens is equipped for navigation.

3 EVALUATION

Dataset: With the Hololens, we collected an in-house dataset, namely **Temmped** (Temporal-Memory-Guided Machine Perception Dataset), due to the lack of existing benchmark datasets. Temmped includes recordings from an infrared (IR) camera that perceives the depth and PV camera that perceives the RGB. We collected 4 egocentric videos from Hololens recording 4 different scenarios for evaluation purposes.

Visualized Temporal-Memory-Guided Machine Perception: A visualized example can be found in Fig. 2. We use the kitchen as an example to verify the practical value of the proposed perception way. We use 2D visualization to show the results. We sample frames at every 0.25 seconds on timestamp, and the 8 frames cover a time span of 2 seconds. In this example, several advantages of memory guidance need to be claimed. First, the history of each object could be recorded in STEMAR, and will give potential guidance to users for their actions. For example, if we have seen the bottle before, potential guidance is cleaning the bottle or drinking water. This cannot be achieved in detection or traditional perception. Second, the location will be recorded. Without the STEMAR, the two bottles

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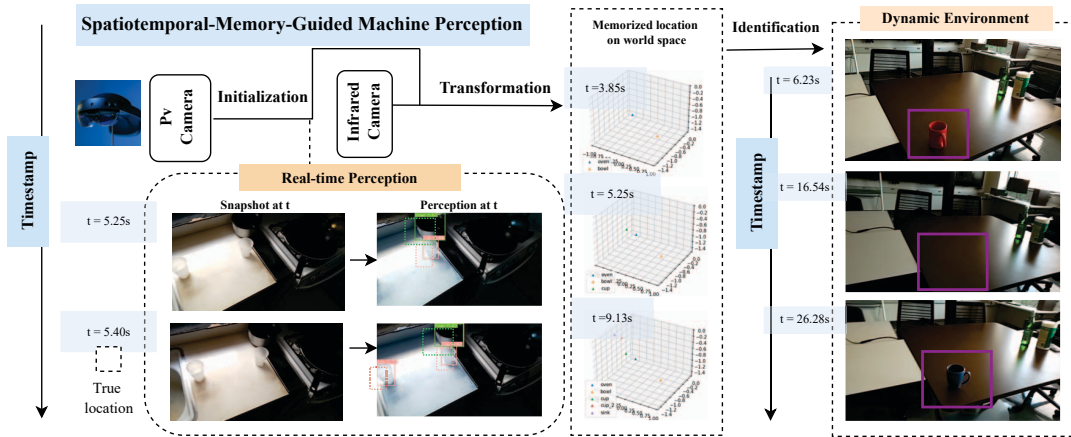


Figure 1: Top: An overview of our method on Hololens. Bottom left: The challenge of latency in real-time perception. The system obtains a snapshot of the world at $t=5.25s$, but when the algorithm completes perception, the surroundings have already changed to $t=5.40s$ on the timestamp. Right: In the proposed dynamic environment task, we can find the changes on the timeline at the location of the pink box: red bottle $\rightarrow$ vacant $\rightarrow$ blue bottle.

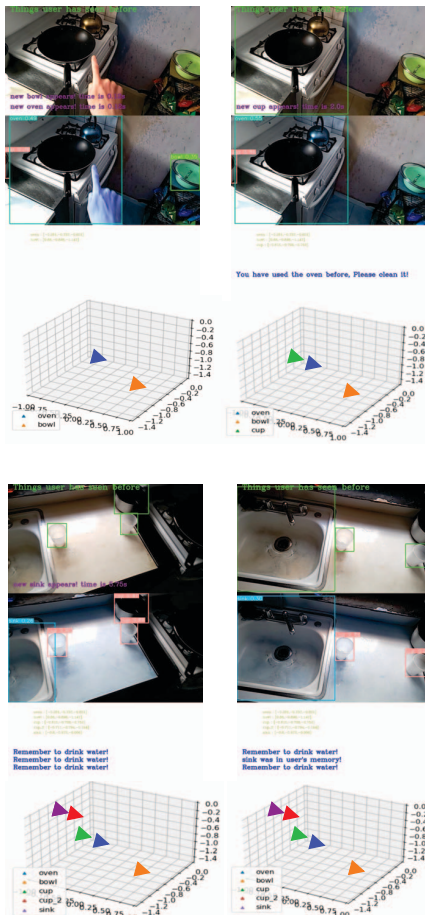


Figure 2: Visualization of Temporal-memory-guided machine perception for Augmented Reality. The first row is a visualization of things in memory and new things; The second row is visualized perception results; the third row is an instruction based on the memory; the last row is the objects visualized in the world space.

with the same appearance and similar surroundings cannot be recognized. The only way to recognize them is through their location in the world space, and STEMAR can make it. Third, the objects can be recognized when the user sees them again after a long time span. We can find in Fig. 2 that everything which is detected in the first frames, can be recaptured in the last several frames. Fourth, the average number of objects that can be seen through a PV camera is no more than 5, within the very limited FoV, but the STEMAR can always hold a world space view.

There's also a side advantage for STEMAR. As we keep the dynamic environment in memory, the memory can give guidance to humans for their actions, as in Fig. 2. It will help with the **knowledge transfer**, from vision knowledge to language knowledge. For example, if we have seen the bottle, when we see it again, it is quite likely to give the user the guidance "wash the bottle".

4 CONCLUSION

In this paper, we propose a novel spatiotemporal-memory-guided machine perception tailored for AR. The presented method mimics the spatiotemporally integrated human memory, which is then leveraged to enhance machine vision accuracy, speed, and robustness. We contextualized the proposed framework to an off-shelf Microsoft Hololens 2 with our real-user data, and verified its effectiveness and practical value.

REFERENCES

- [1] R. E. Clark and S. J. Martin. Interrogating rodents regarding their object and spatial memory. *Current opinion in neurobiology*, 15(5):593–598, 2005.
- [2] S. Kundu, S. Banerjee, A. Raha, S. Natarajan, and K. Basu. Toward functional safety of systolic array-based deep learning hardware accelerators. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 29(3):485–498, 2021.
- [3] M. Li, Y.-X. Wang, and D. Ramanan. Towards streaming perception. In *European Conference on Computer Vision*, pp. 473–488. Springer, 2020.
- [4] X. Luo. From augmented reality to augmented computing: A look at cloud-mobile convergence. In *2009 International Symposium on Ubiquitous Virtual Reality*, pp. 29–32. IEEE, 2009.
- [5] J. Yang, S. Liu, Z. Li, X. Li, and J. Sun. Real-time object detection for streaming perception. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 5385–5395, 2022.